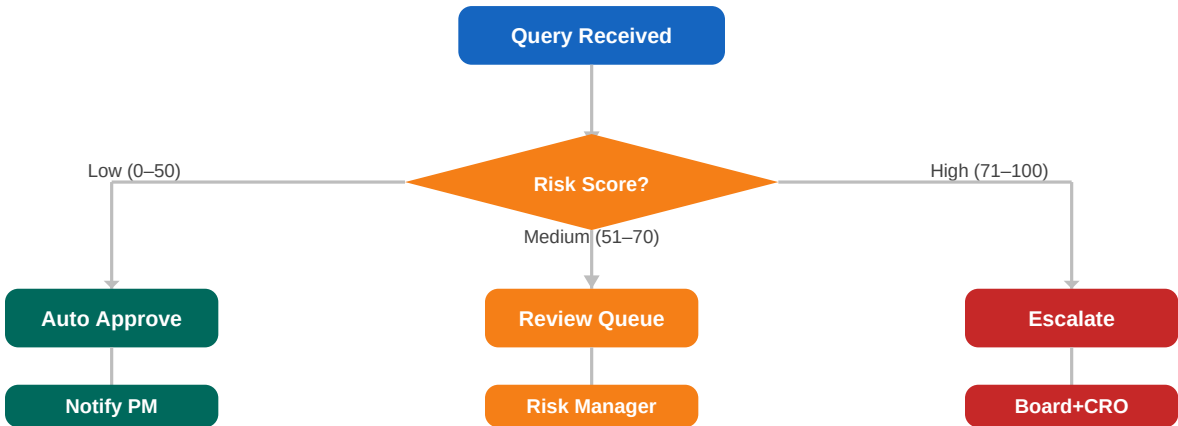


Investment Risk Assessment Framework (Excerpt)

Finance Risk & Investment Intelligence System Doc
Ref: FRIIS-DOC-007 | v1.0 | 2025

This framework defines the risk scoring methodology, classification tiers, investment approval thresholds, and escalation triggers used by the Finance Risk & Investment Intelligence System. All investment decisions must be validated against this framework before execution.

Investment Risk Decision Flow



1. Risk Scoring Methodology

Every asset is assigned a composite risk score from 0 to 100, computed as a weighted average of five risk dimensions evaluated by the system's analytical agents. Scores are recalculated daily:

Risk Dimension	Weight	Data Source	Scoring Method	Max Score
Market Volatility	30%	Market data feed / VIX	Historical vol percentile rank (1-yr)	30
Credit Quality	25%	Rating agency / internal model	Rating-to-score mapping (AAA=2, D=25)	25
Liquidity Profile	20%	Exchange volume / bid-ask spread	Bid-ask spread + average daily volume	20
Regulatory Exposure	15%	Compliance database / jurisdiction flags	Compliance risk matrix × exposure %	15
Concentration Risk	10%	Portfolio holdings database	Herfindahl-Hirschman Index normalised	10

Formula: Composite Score = (Market Vol × 0.30) + (Credit × 0.25) + (Liquidity × 0.20) + (Regulatory × 0.15) + (Concentration × 0.10). Score is recalculated daily and stored in `asset_risk_logs.risk_score`.

2. Risk Classification Tiers

The composite risk score maps to five classification tiers. Each tier triggers a specific monitoring regime, approval requirement, and position size constraint:

Tier	Score Range	Risk Level	Colour Code	Monitoring Regime	Max Position Size
1	0 – 25	Low	Green	Standard quarterly review	12% of portfolio
2	26 – 45	Moderate	Yellow	Enhanced monthly monitoring	10% of portfolio
3	46 – 65	Medium	Amber	Active monitoring + committee review	8% of portfolio
4	66 – 80	High	Orange	Weekly monitoring + escalation flag set	5% of portfolio
5	81 – 100	Critical	Red	Daily monitoring + mandatory escalation	2% of portfolio (hard cap)

3. Investment Approval Thresholds

All investment decisions must be approved at the appropriate governance level based on both the asset risk score and the proposed investment size. The full approval matrix is:

Risk Score	Investment Size	Required Approver	SLA	Documentation Required	System Flag
0–25	Any amount	Portfolio Manager	Same day	Trade ticket; brief rationale	None
26–45	< \$5M	Portfolio Manager	24 hours	Risk note; asset summary brief	Yellow flag
26–45	> \$5M	PM + Risk Manager	48 hours	Full risk assessment report	Yellow flag
46–65	Any amount	Investment Committee	3 business days	Risk assessment + stress test analysis	Amber flag
66–80	Any amount	CRO + Inv. Committee	5 business days	Board risk memo + legal review	Orange flag
81–100	Any amount	Board Approval or Reject	10 business days	Full regulatory review + external audit	Red flag — escalate

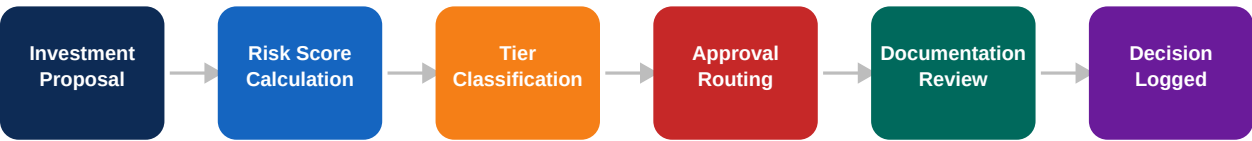
4. Escalation Triggers for High-Risk Assets

The following conditions automatically set `escalation_flag = TRUE` in `asset_risk_logs` and require mandatory human analyst review before any investment decision proceeds. These cannot be overridden by the AI system:

- **Score Threshold Breach:** Any asset with `risk_score > 70` triggers immediate risk manager notification within 2 hours.
- **Database Flag Set:** `escalation_flag = TRUE` in `asset_risk_logs` requires mandatory human analyst review — AI cannot override.
- **Rapid Score Increase:** Two consecutive daily risk score increases of `>10` points trigger enhanced due diligence requirements.
- **Restricted Jurisdiction:** Any asset with exposure to a sanctioned, restricted, or high-risk jurisdiction auto-escalates to Compliance.
- **Active Investigation:** Assets linked to counterparties under regulatory investigation are immediately escalated to Legal and Compliance.
- **Concentration Breach:** Any asset causing portfolio concentration `> 12%` of NAV triggers escalation regardless of individual risk score.
- **Credit Downgrade:** Downgrades below investment grade (BBB-) trigger escalation and a 30-day mandatory review window.
- **Liquidity Deterioration:** Assets where bid-ask spread increases `>200%` week-over-week are flagged for immediate liquidity risk review.

5. Risk Assessment Workflow for New Investments

All new investment proposals must follow this structured workflow before approval and recording in the `investment_decisions` table:



- **Step 1:** Investment proposal submitted with asset name, target portfolio, proposed size, and preliminary investment rationale.
- **Step 2:** System calculates composite risk score across 5 dimensions using live market data, credit data, and portfolio holding data.
- **Step 3:** Asset assigned to risk tier (1–5); `escalation_flag` set if `score > 70` or any critical trigger condition is detected.
- **Step 4:** Approval routing triggered automatically based on risk score × investment size matrix (see Section 3 above).
- **Step 5:** Required documentation reviewed by appointed approver; additional analysis may be requested before a decision is made.
- **Step 6:** Final decision (Approve / Reject / Defer) recorded in `investment_decisions` table with full audit trail and timestamp.

6. Risk Assessment Data Fields Reference

The following fields are central to the risk assessment framework. RAG queries and SQL lookups must reference this schema for accurate results:

Table	Field	Data Type	Description	Query Type
asset_risk_logs	risk_score	INTEGER (0–100)	Composite risk score computed daily by scoring engine	SQL
asset_risk_logs	risk_level	VARCHAR(50)	Tier label: Low / Moderate / Medium / High / Critical	RAG
asset_risk_logs	escalation_flag	BOOLEAN	TRUE if mandatory human review is required	SQL filter
asset_risk_logs	risk_description	TEXT	Narrative description of current risk drivers	RAG — vector search

investment_decisions	approval_status	VARCHAR(50)	Approved / Under Review / Rejected values	SQL lookup
investment_decisions	decision_reason	TEXT	Full rationale for the investment decision made	RAG — vector search
portfolio_holdings	allocation_percentage	FLOAT	Percentage of total portfolio in this holding	SQL + Hybrid
market_indicators	trend_direction	VARCHAR(50)	Rising / Stable / Falling directional label	SQL + Hybrid

■ **AI System Constraint:** The FRIIS AI system must NEVER override an `escalation_flag = TRUE` condition or automatically approve a Tier 4 or Tier 5 (Critical) investment. All such cases require confirmed human analyst sign-off before any decision is executed or logged.

■ Synthetic educational data. All thresholds are illustrative only. Not real investment guidance.